PSO – vzorce

F(x) = p(X < x) · No(μ, σ) · q = 1 − p · kvantily a Φ: tabulky.pdf (zvlášť)

Jevy a kombinatorika

$$p(\bar{A}) = 1 - p(A)$$

$$p(A \cup B) = p(A) + p(B) - p(A \cap B)$$

$$p(A \mid B) = \frac{p(A \cap B)}{p(B)}$$

nezávislé: $p(A \cap B) = p(A)\,p(B)$

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}, \qquad V_k(n) = \frac{n!}{(n-k)!}, \qquad V'_k(n) = n^k$$

$$p(A) = \frac{m}{n}, \qquad p(A) = \frac{|A|}{|\Omega|} \text{ (geometrická: délka, plocha)}$$

Úplná pravděpodobnost, Bayes, opakování

$$p(A) = \sum_i p(H_i)\,p(A \mid H_i), \qquad \sum_i p(H_i) = 1$$

$$p(H_k \mid A) = \frac{p(H_k)\,p(A \mid H_k)}{\sum_i p(H_i)\,p(A \mid H_i)}$$

$$p(A \text{ dřív než } B) = \frac{p(A)}{p(A) + p(B)}$$

$$p(\text{začínající vyhraje}) = \frac{a}{1 - (1-a)(1-b)}$$

$$\sum_{k=0}^{\infty} q^k = \frac{1}{1-q}, \qquad \sum_{k=0}^{\infty} \frac{x^k}{k!} = e^x$$

Náhodná veličina

$$F(x) = p(X < x), \qquad p(a \le X < b) = F(b) - F(a)$$

$$p(X = x_i) = F(x_i^+) - F(x_i)$$

$$EX = \sum x_i p_i \quad \text{nebo} \quad \int x f(x)\,dx$$

$$EX^2 = \sum x_i^2 p_i \quad \text{nebo} \quad \int x^2 f(x)\,dx$$

$$DX = EX^2 - (EX)^2, \qquad \sigma = \sqrt{DX}$$

$$E(aX + b) = a\,EX + b, \qquad D(aX + b) = a^2 DX$$

$$E(X + Y) = EX + EY$$

nezávislé: $D(X + Y) = DX + DY$

$$\int_{-\infty}^{\infty} f(x)\,dx = 1, \qquad f \ge 0, \qquad F(x) = \int_{-\infty}^{x} f(t)\,dt$$

$$p(a < X < b) = \int_a^b f(x)\,dx, \qquad F(\tilde{x}) = 0{,}5$$

Diskrétní rozdělení

	$p(X = k)$	EX	DX
Bi(n, p), k = 0..n	$\binom{n}{k} p^k q^{n-k}$	np	npq
Ge, k = 0, 1, …	$p^k q$	$\frac{p}{q}$	$\frac{p}{q^2}$
Ge, k = 1, 2, …	$q^{k-1} p$	$\frac{1}{p}$	$\frac{q}{p^2}$
H(N, M, n)	$\frac{\binom{M}{k}\binom{N-M}{n-k}}{\binom{N}{n}}$	$n\frac{M}{N}$	$n\frac{M}{N}\frac{N-M}{N}\frac{N-n}{N-1}$
Po(λ), k = 0, 1, …	$\frac{\lambda^k}{k!} e^{-\lambda}$	λ	λ

Bi(n, p) ≈ Po(np) pro velké n, malé p.

Spojitá rozdělení

	$f(x)$	EX	DX
rovnoměrné ⟨a, b⟩	$\frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
exponenciální, x ≥ 0	$\lambda e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Laplace	$\frac{\lambda}{2} e^{-\lambda \lvert x \rvert}$	0	$\frac{2}{\lambda^2}$
No(μ, σ)	$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2

Integrály

$$\int x^n\,dx = \frac{x^{n+1}}{n+1}, \qquad \int e^{ax} dx = \frac{e^{ax}}{a}$$

$$\int \frac{dx}{x} = \ln|x|$$

$$\int uv'\,dx = uv - \int u'v\,dx$$

$$\int x \sin x\,dx = \sin x - x\cos x$$

$$\int x \cos x\,dx = \cos x + x \sin x$$

$$\int x^2 \sin x\,dx = -x^2 \cos x + 2x \sin x + 2\cos x$$

$$\int x^2 \cos x\,dx = x^2 \sin x + 2x \cos x - 2 \sin x$$

$$\int x e^{-\lambda x} dx = -\frac{e^{-\lambda x}}{\lambda^2}(\lambda x + 1)$$

$$\int x^2 e^{-\lambda x} dx = -\frac{e^{-\lambda x}}{\lambda^3}(\lambda^2 x^2 + 2\lambda x + 2)$$

Normální rozdělení

$$u = \frac{x - \mu}{\sigma}, \qquad p(X < x) = \Phi(u)$$

$$\Phi(-u) = 1 - \Phi(u)$$

$$p(a < X < b) = \Phi\left(\tfrac{b-\mu}{\sigma}\right) - \Phi\left(\tfrac{a-\mu}{\sigma}\right)$$

$$x_p = \mu + \sigma u_p, \qquad u_{1-p} = -u_p$$

$$p(|X - \mu| < \sigma u_{(1+\gamma)/2}) = \gamma$$

Limitní věty

$$\mathrm{Bi}(n, p) \approx \mathrm{No}\left(np,\ \sqrt{npq}\right), \qquad npq > 9$$

$$\tfrac{X}{n} \approx \mathrm{No}\left(p,\ \sqrt{pq/n}\right)$$

$$\textstyle\sum_{i=1}^{n} X_i \approx \mathrm{No}\left(n\mu,\ \sigma\sqrt{n}\right)$$

$$\bar{X} \approx \mathrm{No}\left(\mu,\ \sigma/\sqrt{n}\right)$$

$$p(|X - EX| \ge \varepsilon) \le \frac{DX}{\varepsilon^2}$$

Náhodný vektor

$$\iint f(x,y)\,dx\,dy = 1$$

$$p((X,Y)\in G) = \iint_G f(x,y)\,dx\,dy$$

$$f_X(x) = \int f(x,y)\,dy, \qquad f_Y(y) = \int f(x,y)\,dx$$

$$\text{nezávislé} \iff f(x,y) = f_X(x)\,f_Y(y)$$

$$\operatorname{cov}(X,Y) = E(XY) - EX\,EY$$

$$\rho = \frac{\operatorname{cov}(X,Y)}{\sigma_X \sigma_Y}$$

Výběrové charakteristiky

$$\bar{x} = \frac{1}{n}\sum x_i$$

$$s^2 = \frac{1}{n-1}\sum (x_i-\bar{x})^2 = \frac{\sum x_i^2 - (\sum x_i)^2/n}{n-1}$$

četnosti: $\bar{x} = \dfrac{\sum x_j n_j}{n}, \qquad \sigma^2 = \dfrac{\sum x_j^2 n_j}{n} - \bar{x}^2$

Intervaly spolehlivosti (1 − α)

$$\mu \in \bar{x} \pm t_{1-\alpha/2}(n-1)\,\frac{s}{\sqrt{n}}$$

$$\mu \in \bar{x} \pm u_{1-\alpha/2}\,\frac{\sigma}{\sqrt{n}} \quad (\sigma \text{ známé})$$

$$\sigma^2 \in \left(\frac{(n-1)s^2}{\chi^2_{1-\alpha/2}(n-1)},\ \frac{(n-1)s^2}{\chi^2_{\alpha/2}(n-1)} \right)$$

Testy (K1–K5)

Oboustranný test: kvantil $1-\alpha/2$; jednostranný: $1-\alpha$.

$$T = \frac{\bar{x}-\mu_0}{s}\sqrt{n} \sim t(n-1)$$

$$U = \frac{\bar{x}-\mu_0}{\sigma}\sqrt{n} \sim \mathrm{No}(0,1)$$

$$d_i = x_i - y_i: \quad T = \frac{\bar{d}}{s_d}\sqrt{n} \sim t(n-1)$$

$$s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}$$

$$T = \frac{\bar{x}_1-\bar{x}_2}{s_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} \sim t(n_1+n_2-2)$$

$$U = \frac{\bar{x}_1-\bar{x}_2}{\sqrt{\frac{s_1^2}{n_1}+\frac{s_2^2}{n_2}}} \approx \mathrm{No}(0,1)$$

$$F = \frac{s_1^2}{s_2^2} \sim F(n_1-1,\ n_2-1), \qquad s_1^2 \geq s_2^2$$

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2} \sim \chi^2(n-1)$$

Zamítáme H_0: $|T| \geq t_{1-\alpha/2}$; $T \leq -t_{1-\alpha}$; $T \geq t_{1-\alpha}$; $F \geq F_{1-\alpha/2}$.

Testy dobré shody

$$\chi^2 = \sum_{j=1}^{k} \frac{(n_j - E_j)^2}{E_j}, \qquad E_j = n\,p_j \geq 5$$

zamítáme: $\chi^2 \geq \chi^2_{1-\alpha}(k-1-r)$

r = počet odhadnutých parametrů (Po: $\lambda = \bar{x}$, r = 1; No: $\mu = \bar{x}$, σ, r = 2).

kontingenční tabulka: $E_{ij} = \dfrac{r_i\,s_j}{n}, \quad \nu = (r-1)(s-1)$

K–S: $F_n(x_j) = \dfrac{n_1 + \cdots + n_j}{n}$

$$D = \max_j \left| F_n(x_j) - F_0(x_j) \right|$$

zamítáme: $D \geq D_n(\alpha)$

$n > 40$: $D_n(0{,}05) \approx \dfrac{1{,}36}{\sqrt{n}}, \quad D_n(0{,}01) \approx \dfrac{1{,}63}{\sqrt{n}}$

Regrese a korelace

$$\hat{y} = a + bx, \qquad a = \bar{y} - b\,\bar{x}$$

$$b = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}$$

$$r = \frac{n\sum xy - \sum x \sum y}{\sqrt{\left(n\sum x^2 - (\sum x)^2\right)\left(n\sum y^2 - (\sum y)^2\right)}}$$